This R code is to generate the tables and plots for analyzing obesity paradox.

- Description analysis.
- Evaluate the association between obesity status and CPC counts - linear regression.
- Evaluate the association of obesity and CPC counts with cardiovascular outcomes - Cox model.

Table 1: Baseline characteristics of CHDWB participants stratified by obesity status

| | Overall | Non_obese | Obese | p_value |
|---|---|---|---|---|
| age | 50.0 [41.0, 57.0] | 50.0 [40.0, 57.0] | 50.0 [42.5, 56.5] | 0.537 |
| BMI | 26.5 [23.6, 30.6] | 25.0 [22.5, 26.9] | 34.2 [31.6, 37.8] | <0.001 |
| eGFR | 95.1 [86.2, 106.5] | 94.7 [85.5, 104.8] | 96.3 [87.2, 110.4] | 0.123 |
| hsCRPmgL | 1.5 [0.5, 3.5] | 1.0 [0.5, 2.1] | 3.8 [2.0, 6.7] | <0.001 |
| I_HOMA2IR | 0.4 [0.1, 0.9] | 0.3 [0.1, 0.7] | 1.1 [0.7, 1.7] | <0.001 |
| I_HOMA2B | 58.0 [31.8, 93.7] | 41.4 [29.5, 74.5] | 100.1 [68.8, 132.8] | <0.001 |
| VO2MAX | 33.5 [27.2, 39.2] | 35.4 [29.6, 41.3] | 27.7 [24.3, 33.1] | <0.001 |
| BFP | 36.0 [29.0, 41.8] | 32.2 [27.1, 38.3] | 44.6 [39.3, 49.0] | <0.001 |
| AFP | 3.3 [2.6, 4.0] | 2.9 [2.2, 3.5] | 4.4 [3.9, 5.0] | <0.001 |
| GFP | 7.3 [5.5, 8.6] | 7.0 [5.1, 8.3] | 8.3 [6.7, 9.4] | <0.001 |
| wbc_count | 5.5 [4.7, 6.7] | 5.4 [4.6, 6.4] | 6.1 [5.2, 7.3] | <0.001 |
| CD34 | 2115.2 [1327.6, 3124.3] | 1990.4 [1277.4, 2850.0] | 2431.5 [1574.7, 3939.2] | <0.001 |
| CD133 | 889.4 [553.9, 1397.1] | 799.7 [532.9, 1268.3] | 1147.5 [722.1, 1795.4] | <0.001 |
| CXCR4 | 858.1 [524.0, 1381.1] | 780.9 [509.7, 1280.1] | 985.8 [572.0, 1551.7] | 0.002 |
| male | 230 (34.2) | 180 (37.4) | 50 (26.2) | 0.007 |
| Black | 151 (22.5) | 78 (16.2) | 73 (38.2) | <0.001 |
| DM | 75 (11.2) | 34 (7.1) | 41 (21.5) | <0.001 |
| HTN | 229 (34.1) | 131 (27.2) | 98 (51.3) | <0.001 |
| curr_smoker | 36 (5.4) | 22 (4.6) | 14 (7.3) | 0.219 |
| HC | 110 (16.4) | 63 (13.1) | 47 (24.6) | <0.001 |

Table 2: Association of obesity with circulating progenitor cells counts in CHDWB cohort

| | CD34 | | | | CD133 | | | | CXCR4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Estimated difference | 2.5%CI | 97.5%CI | p-value | Estimated difference | 2.5%CI | 97.5%CI | p-value | Estimated difference | 2.5%CI | 97.5%CI | p-value |
| Univariate | 28.2 | 14.0 | 44.1 | <0.001 | 43.1 | 25.4 | 63.2 | <0.001 | 21.0 | 6.5 | 37.4 | 0.003 |
| Clinical Model | 21.6 | 7.9 | 37.1 | 0.001 | 34.2 | 17.1 | 53.6 | <0.001 | 16.4 | 2.0 | 32.8 | 0.024 |
| **model1** | | | | | | | | | | | | |
| model1_Obese | 20.7 | 5.1 | 38.7 | 0.008 | 32.5 | 13.1 | 55.3 | 0.001 | 19.3 | 2.6 | 38.7 | 0.022 |
| HsCRP | -3.1 | -9.2 | 3.4 | 0.336 | -3.4 | -10.3 | 4.1 | 0.367 | -5.5 | -11.9 | 1.4 | 0.117 |
| **model2** | | | | | | | | | | | | |
| model2_Obese | 22.7 | 7.4 | 40.1 | 0.003 | 32.9 | 14.3 | 54.5 | <0.001 | 19.6 | 3.3 | 38.4 | 0.017 |
| HOMA2IR | -0.9 | -6.6 | 5.2 | 0.765 | 1.0 | -5.6 | 8.0 | 0.776 | -2.7 | -8.9 | 3.8 | 0.405 |
| **model3** | | | | | | | | | | | | |
| model3_Obese | 21.3 | 6.6 | 38.0 | 0.003 | 33.5 | 15.3 | 54.5 | <0.001 | 17.9 | 2.3 | 36.0 | 0.023 |
| HOMA2 | 0.3 | -5.3 | 6.2 | 0.923 | 0.6 | -5.7 | 7.4 | 0.854 | -1.6 | -7.6 | 4.9 | 0.621 |
| **model4** | | | | | | | | | | | | |
| model4_Obese | 13.3 | -0.5 | 29.1 | 0.059 | 29.3 | 11.4 | 50.2 | 0.001 | 8.0 | -6.4 | 24.5 | 0.294 |
| VO2 | -3.5 | -9.3 | 2.6 | 0.255 | -3.0 | -9.7 | 4.1 | 0.398 | -3.6 | -10.0 | 3.1 | 0.285 |
| **model5** | | | | | | | | | | | | |
| model5_Obese | 21.5 | 6.6 | 38.6 | 0.004 | 33.5 | 15.0 | 55.1 | <0.001 | 18.6 | 2.5 | 37.3 | 0.022 |
| GFP | 3.4 | -5.5 | 13.1 | 0.466 | 8.6 | -2.0 | 20.3 | 0.116 | -2.2 | -11.5 | 8.1 | 0.659 |
| **model6** | | | | | | | | | | | | |
| model6_Obese | 14.1 | -2.3 | 33.2 | 0.096 | 26.2 | 5.7 | 50.6 | 0.010 | 8.2 | -8.9 | 28.6 | 0.368 |
| AFP | 6.3 | -1.1 | 14.3 | 0.095 | 7.8 | -0.7 | 17.1 | 0.075 | 6.4 | -1.8 | 15.3 | 0.131 |
| **model7** | | | | | | | | | | | | |
| model7_Obese | 15.3 | 0.7 | 32.0 | 0.040 | 30.2 | 11.5 | 52.0 | 0.001 | 8.4 | -6.8 | 26.0 | 0.295 |
| AFP2 | 9.1 | 1.5 | 17.2 | 0.018 | 8.7 | 0.1 | 18.0 | 0.046 | 10.4 | 2.0 | 19.6 | 0.015 |
| **model8** | | | | | | | | | | | | |
| model8_Obese | 23.4 | 8.0 | 40.9 | 0.002 | 41.8 | 21.9 | 65.1 | <0.001 | 15.6 | -0.3 | 34.0 | 0.054 |
| GFP2 | -0.2 | -7.4 | 7.5 | 0.956 | 3.0 | -5.4 | 12.2 | 0.489 | -2.2 | -10.0 | 6.3 | 0.599 |
| **model9** | | | | | | | | | | | | |
| model9_Obese | 13.5 | -2.5 | 32.3 | 0.103 | 22.4 | 2.9 | 45.7 | 0.023 | 13.4 | -4.3 | 34.3 | 0.146 |
| Lean_pct | -0.5 | -1.5 | 0.4 | 0.269 | -0.7 | -1.8 | 0.3 | 0.181 | -0.2 | -1.2 | 0.9 | 0.776 |

[a] Z-scores were created for VO2, AFP, AFP2, GFP and GFP2

Table 3: Association of obesity and CPC counts with cardiovascular outcomes in EmCAB cohort

| | Cardiovascular Death/MI | | | | All-Cause Death | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | 2.5%CI | 97.5%CI | p-value | HR | 2.5%CI | 97.5%CI | p-value |
| Univariate_Obese | 0.678 | 0.491 | 0.935 | 0.018 | 0.652 | 0.492 | 0.863 | 0.003 |
| clinical_Obese | 0.704 | 0.500 | 0.992 | 0.045 | 0.721 | 0.536 | 0.971 | 0.032 |
| **model1** | | | | | | | | |
| Obese | 0.729 | 0.517 | 1.027 | 0.071 | 0.744 | 0.552 | 1.003 | 0.053 |
| Log0.5_CD34 | 1.299 | 1.126 | 1.498 | <0.001 | 1.267 | 1.119 | 1.434 | <0.001 |
| **model2** | | | | | | | | |
| Obese | 0.719 | 0.510 | 1.013 | 0.059 | 0.738 | 0.547 | 0.994 | 0.046 |
| Log0.5_CD133 | 1.217 | 1.100 | 1.346 | <0.001 | 1.185 | 1.081 | 1.299 | <0.001 |
| **model3** | | | | | | | | |
| Obese | 0.718 | 0.510 | 1.012 | 0.058 | 0.738 | 0.548 | 0.995 | 0.046 |
| Log0.5_CXCR4 | 1.323 | 1.172 | 1.493 | <0.001 | 1.256 | 1.121 | 1.408 | <0.001 |

Table 4: Cardiovascular outcomes in EmCAB cohort participants stratified by Obesity and CPC counts

| | Cardiovascular Death/MI | | | | All-Cause Death | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | 2.5%CI | 97.5%CI | p-value | HR | 2.5%CI | 97.5%CI | p-value |
| **CD34** | | | | | | | | |
| Obese and High CD34+ | 0.522 | 0.310 | 0.876 | 0.014 | 0.483 | 0.303 | 0.769 | 0.002 |
| Obese and Low CD34+ | 0.745 | 0.484 | 1.146 | 0.180 | 0.802 | 0.553 | 1.162 | 0.243 |
| Non-Obese and High CD34+ | 0.824 | 0.568 | 1.196 | 0.309 | 0.814 | 0.593 | 1.118 | 0.204 |
| **CD133** | | | | | | | | |
| Obese and High CD34+/CD133+ | 0.572 | 0.341 | 0.958 | 0.034 | 0.536 | 0.337 | 0.852 | 0.008 |
| Obese and Low CD34+/CD133+ | 0.770 | 0.498 | 1.191 | 0.241 | 0.816 | 0.561 | 1.185 | 0.285 |
| Non-Obese and High CD34+/CD133+ | 0.933 | 0.646 | 1.348 | 0.713 | 0.913 | 0.667 | 1.251 | 0.572 |
| **CXCR4** | | | | | | | | |
| Obese and High CD34+/CXCR4+ | 0.394 | 0.225 | 0.689 | 0.001 | 0.450 | 0.279 | 0.727 | 0.001 |
| Obese and Low CD34+/CXCR4+ | 0.832 | 0.550 | 1.258 | 0.383 | 0.850 | 0.591 | 1.224 | 0.383 |
| Non-Obese and High CD34+/CXCR4+ | 0.766 | 0.527 | 1.113 | 0.161 | 0.835 | 0.608 | 1.146 | 0.264 |

Table 5: Association of obesity with circulating progenitor cells counts in CHDWB cohort (interaction between obese and sex)

| | CD34 | | | | CD133 | | | | CXCR4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Estimated difference | 2.5%CI | 97.5%CI | p-value | Estimated difference | 2.5%CI | 97.5%CI | p-value | Estimated difference | 2.5%CI | 97.5%CI | p-value |
| **Univariate** | | | | | | | | | | | | |
| Univariate_Obese | 26.1 | 10.0 | 44.6 | 0.001 | 37.3 | 17.6 | 60.2 | <0.001 | 21.5 | 4.7 | 41.0 | 0.010 |
| Obese * Male | 19.2 | -7.5 | 53.7 | 0.175 | 28.4 | -3.7 | 71.2 | 0.088 | 11.6 | -15.4 | 47.2 | 0.437 |
| **Clinical** | | | | | | | | | | | | |
| Clinical_Obese | 15.9 | 0.8 | 33.2 | 0.038 | 25.1 | 6.9 | 46.4 | 0.005 | 13.2 | -2.9 | 32.0 | 0.113 |
| Obese * Male1 | 18.1 | -7.6 | 50.8 | 0.183 | 27.2 | -3.6 | 67.8 | 0.088 | 10.0 | -16.0 | 44.1 | 0.486 |

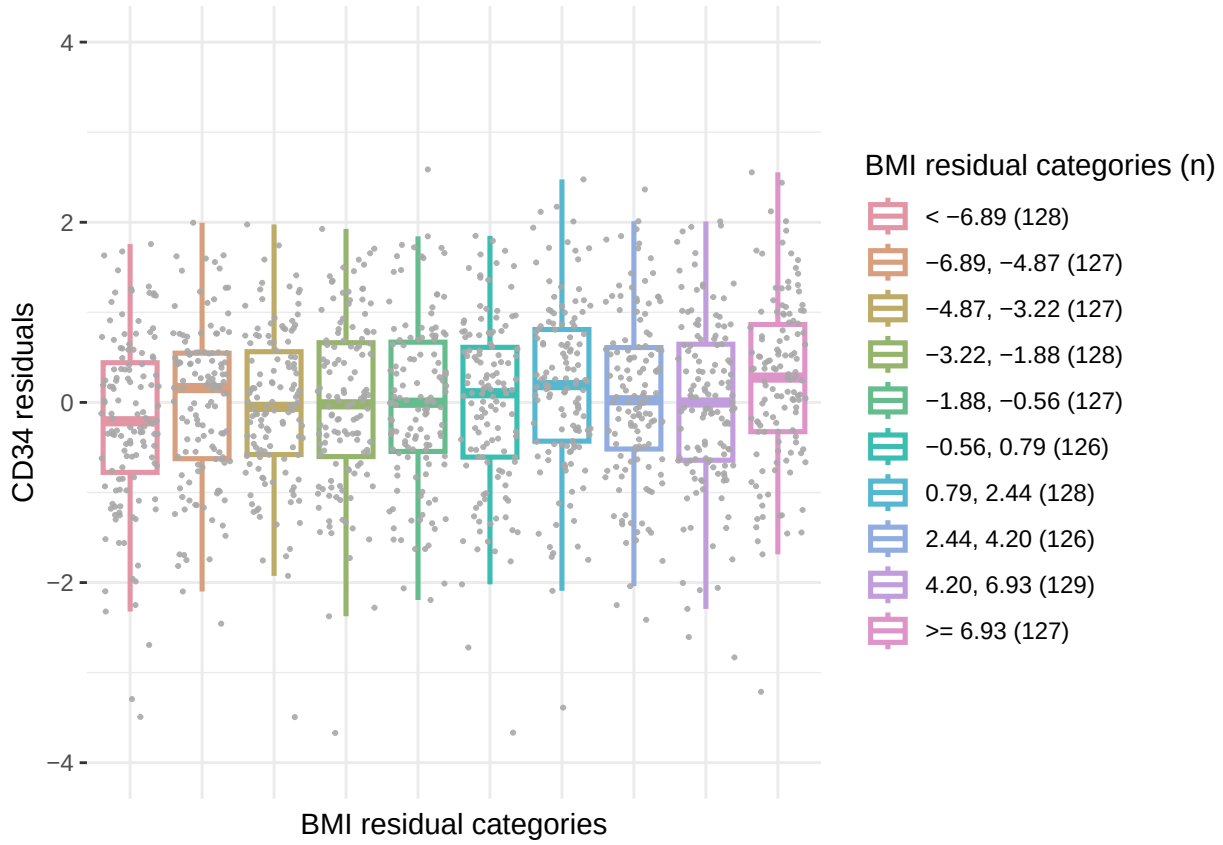
Table 6: Association of obesity and CPC counts with cardiovascular outcomes in EmCAB cohort (interaction between obese and sex)

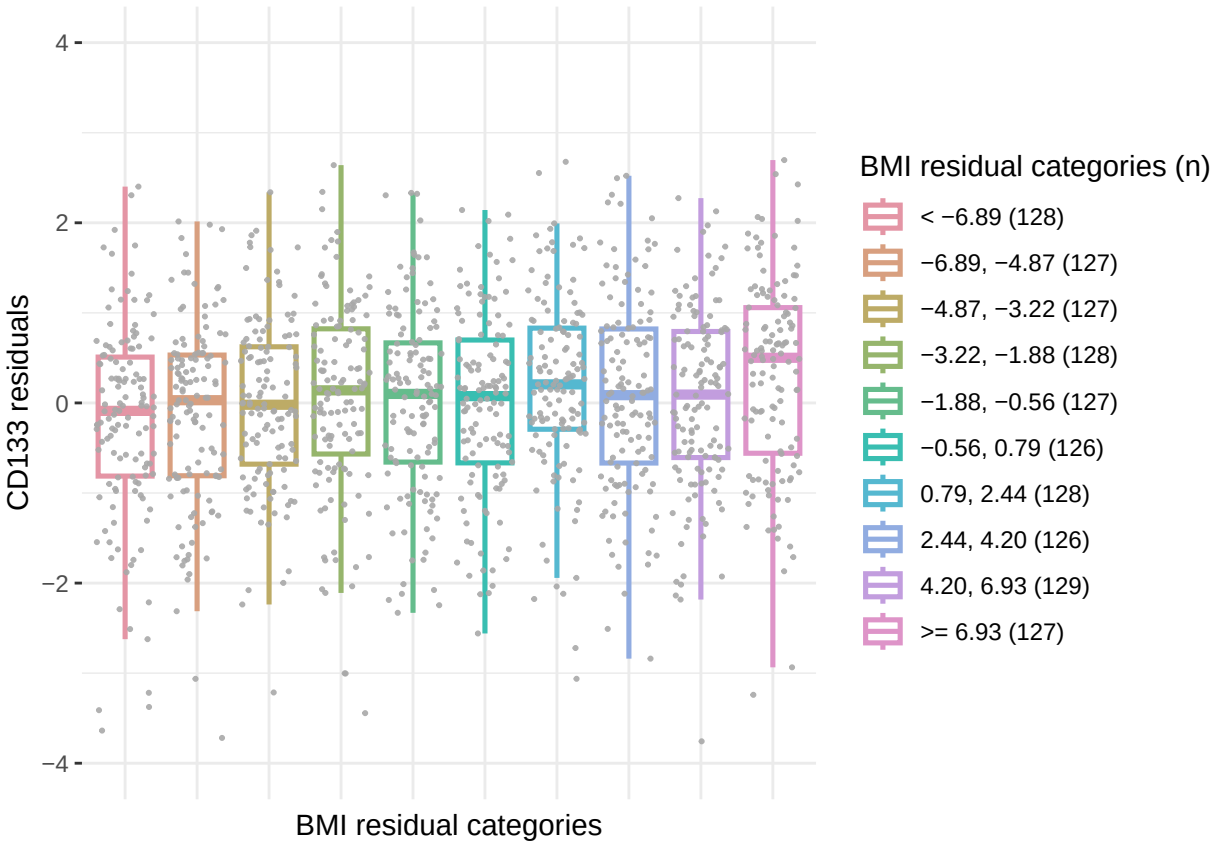
| | Cardiovascular Death/MI | | | | All-Cause Death | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | 2.5%CI | 97.5%CI | p-value | HR | 2.5%CI | 97.5%CI | p-value |
| **Univariate** | | | | | | | | |
| Univariate_Obese | 0.952 | 0.593 | 1.529 | 0.840 | 0.810 | 0.534 | 1.229 | 0.322 |
| Obese * male | 0.538 | 0.280 | 1.034 | 0.063 | 0.677 | 0.384 | 1.193 | 0.177 |
| **Clinical** | | | | | | | | |
| Clinical_Obese | 0.955 | 0.586 | 1.558 | 0.854 | 0.864 | 0.562 | 1.328 | 0.505 |
| Obese * male | 0.572 | 0.295 | 1.108 | 0.098 | 0.722 | 0.408 | 1.280 | 0.265 |
| **model1** | | | | | | | | |
| Obese | 0.941 | 0.578 | 1.533 | 0.807 | 0.859 | 0.559 | 1.321 | 0.490 |
| Log0.5_CD34 | 1.289 | 1.117 | 1.487 | 0.001 | 1.263 | 1.115 | 1.430 | <0.001 |
| Obese*male | 0.621 | 0.320 | 1.205 | 0.159 | 0.770 | 0.434 | 1.365 | 0.370 |
| **model2** | | | | | | | | |
| Obese | 0.922 | 0.564 | 1.506 | 0.745 | 0.851 | 0.553 | 1.310 | 0.463 |
| Log0.5_CD133 | 1.206 | 1.090 | 1.335 | <0.001 | 1.180 | 1.077 | 1.294 | <0.001 |
| Obese*male | 0.631 | 0.324 | 1.229 | 0.176 | 0.771 | 0.434 | 1.370 | 0.376 |
| **model3** | | | | | | | | |
| Obese | 0.924 | 0.567 | 1.506 | 0.752 | 0.849 | 0.553 | 1.305 | 0.457 |
| Log0.5_CXCR4 | 1.318 | 1.165 | 1.491 | <0.001 | 1.253 | 1.117 | 1.405 | <0.001 |
| Obese*male | 0.626 | 0.323 | 1.213 | 0.165 | 0.774 | 0.436 | 1.373 | 0.381 |

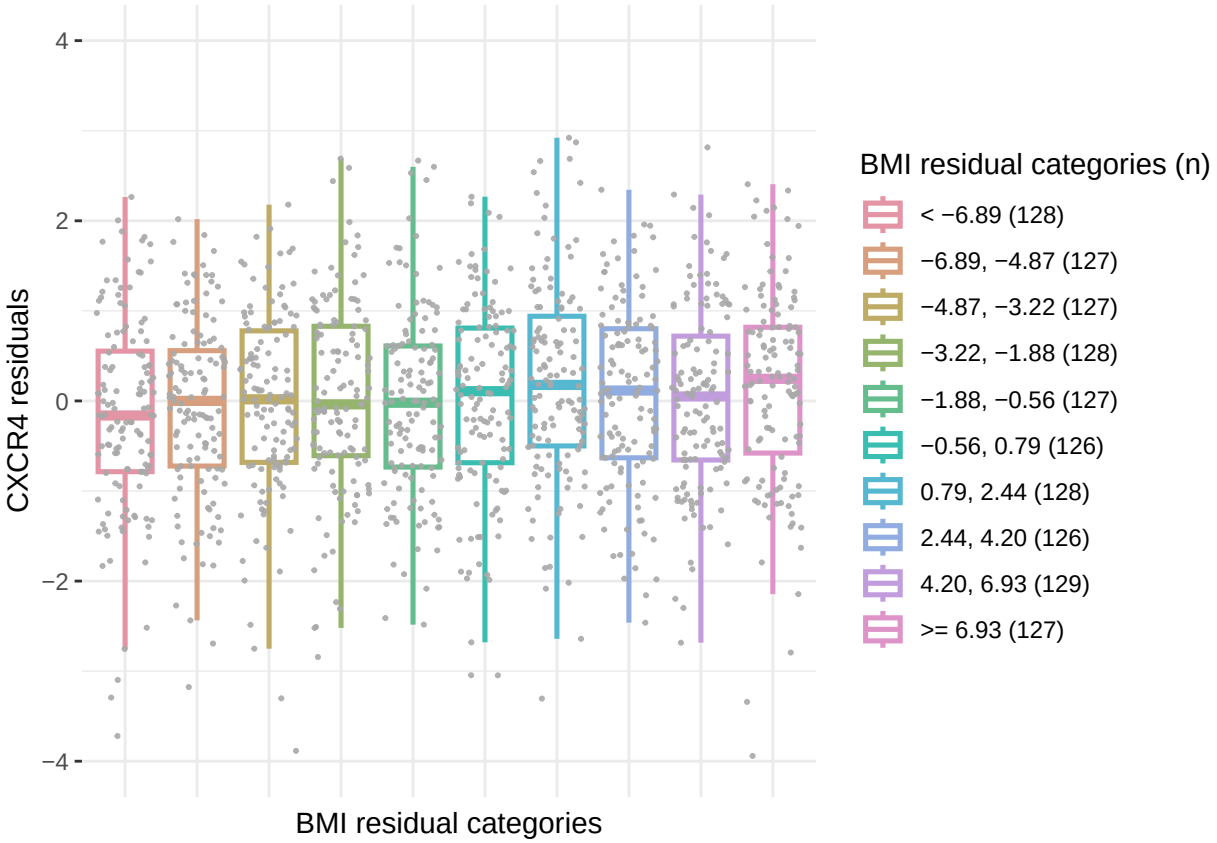
Table 7: Association of obesity and CPC counts with cardiovascular outcomes in EmCAB cohort (interaction between CPC and sex)

| | Cardiovascular Death/MI | | | | All-Cause Death | | | |
|---|---|---|---|---|---|---|---|---|
| | HR | 2.5%CI | 97.5%CI | p-value | HR | 2.5%CI | 97.5%CI | p-value |
| **model1** | | | | | | | | |
| Obese | 0.730 | 0.518 | 1.030 | 0.073 | 0.742 | 0.550 | 1.000 | 0.050 |
| CD34 | 1.279 | 1.017 | 1.609 | 0.035 | 1.308 | 1.077 | 1.590 | 0.007 |
| CD34 * male | 1.025 | 0.766 | 1.372 | 0.867 | 0.948 | 0.738 | 1.218 | 0.679 |
| **model2** | | | | | | | | |
| Obese | 0.710 | 0.503 | 1.004 | 0.053 | 0.732 | 0.543 | 0.988 | 0.042 |
| CD133 | 1.246 | 1.102 | 1.409 | <0.001 | 1.213 | 1.078 | 1.364 | 0.001 |
| CD133 * male | 0.943 | 0.773 | 1.151 | 0.562 | 0.950 | 0.795 | 1.135 | 0.571 |
| **model3** | | | | | | | | |
| Obese | 0.721 | 0.511 | 1.016 | 0.061 | 0.741 | 0.550 | 0.999 | 0.049 |
| CXCR44 | 1.267 | 1.020 | 1.574 | 0.033 | 1.198 | 0.991 | 1.449 | 0.062 |
| CXCR44 * male | 1.064 | 0.822 | 1.377 | 0.637 | 1.076 | 0.851 | 1.359 | 0.541 |

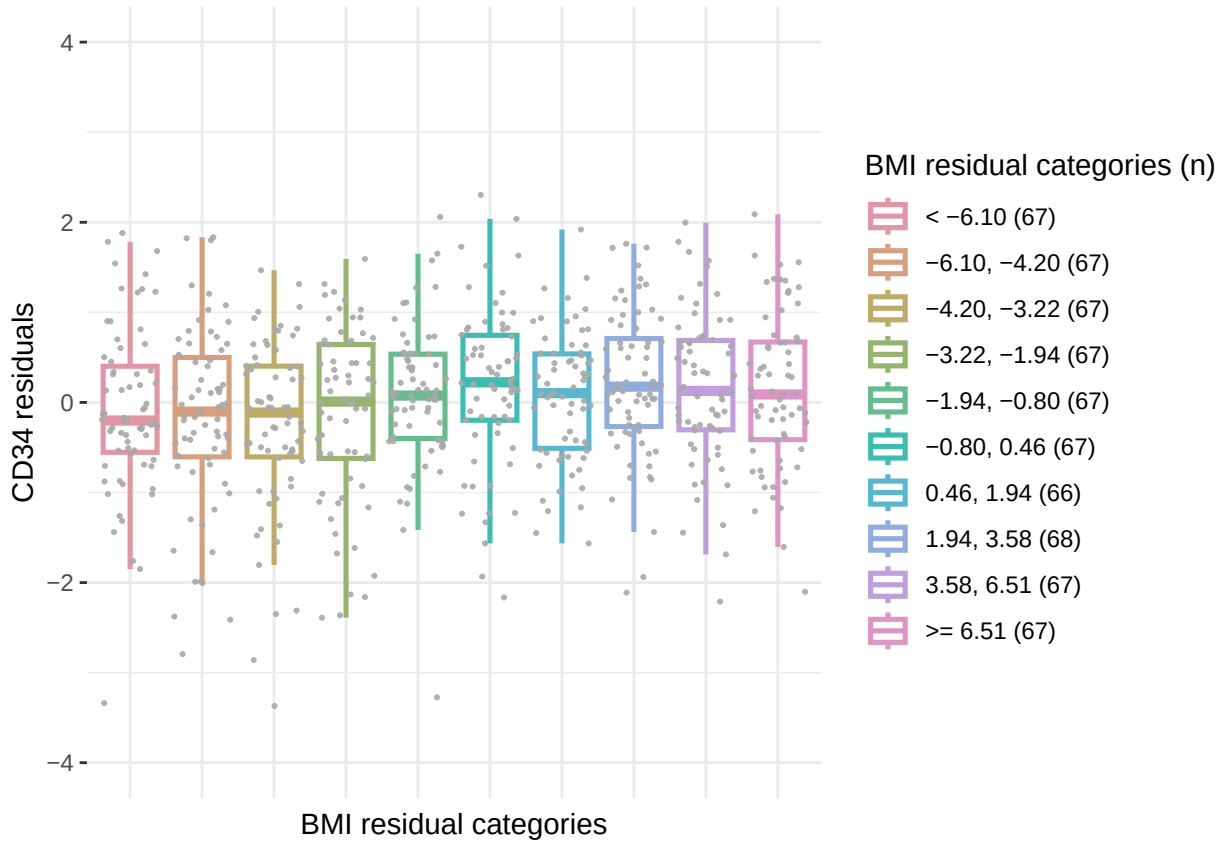
**partial plots using ECB data set**

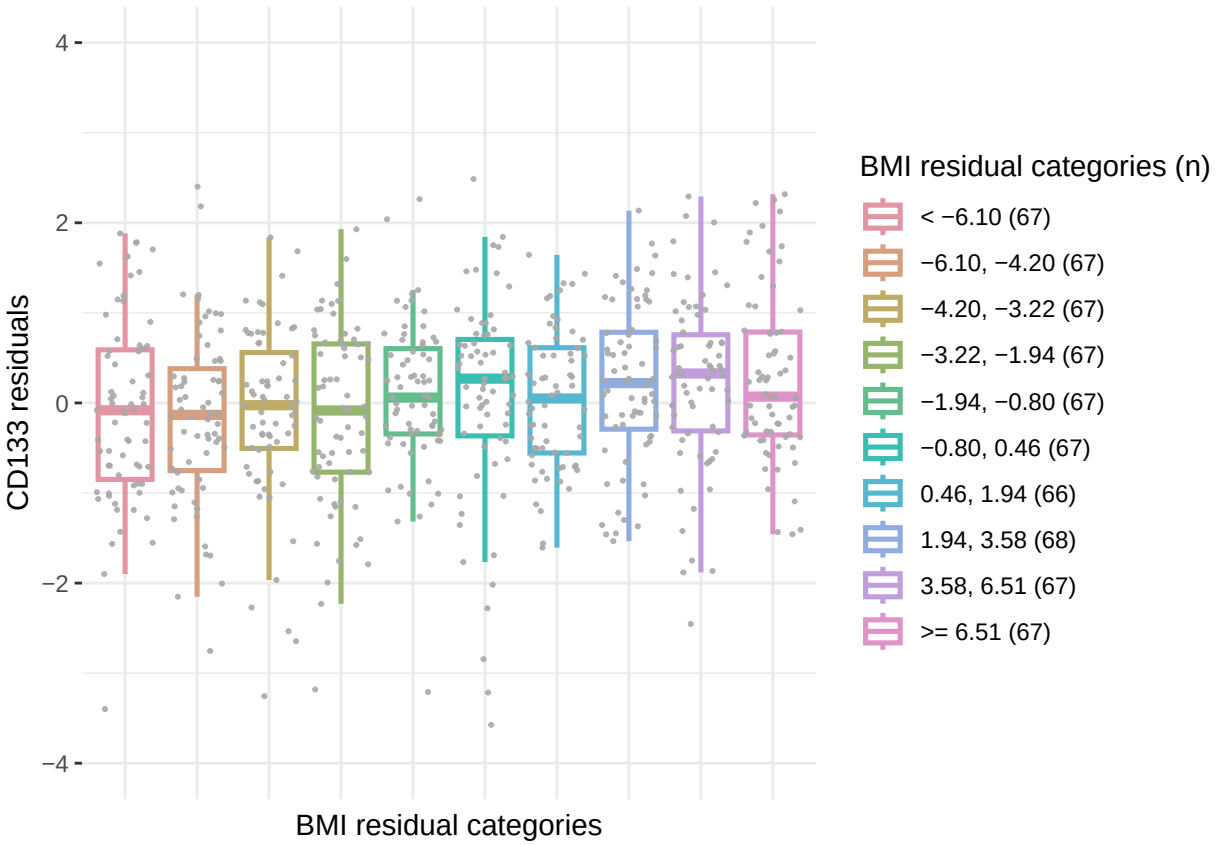


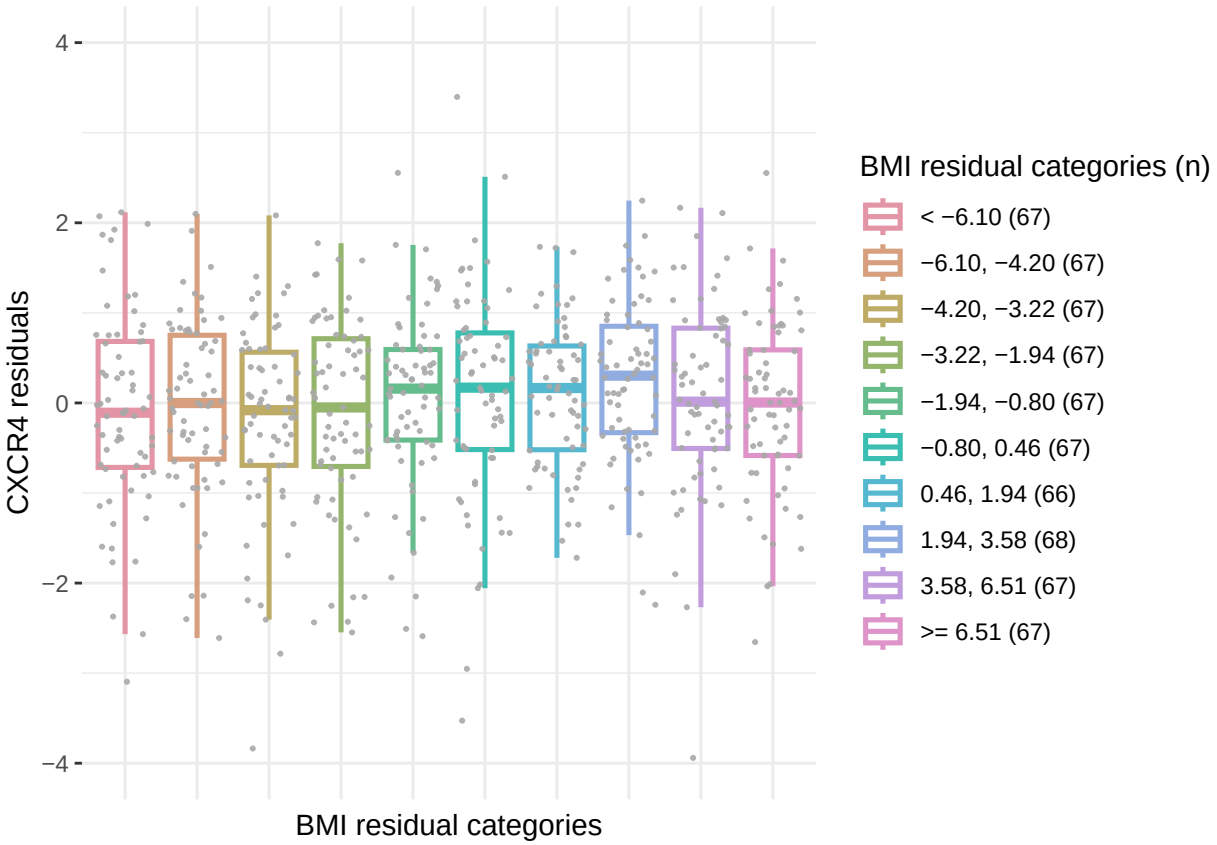

4
2
0
−2
−4
CD133 residuals
BMI residual categories
BMI residual categories (n)
< −6.89 (128)
−6.89, −4.87 (127)
−4.87, −3.22 (127)
−3.22, −1.88 (128)
−1.88, −0.56 (127)
−0.56, 0.79 (126)
0.79, 2.44 (128)
2.44, 4.20 (126)
4.20, 6.93 (129)
>= 6.93 (127)


4
2
0
−2
−4
CXCR4 residuals
BMI residual categories
BMI residual categories (n)
< −6.89 (128)
−6.89, −4.87 (127)
−4.87, −3.22 (127)
−3.22, −1.88 (128)
−1.88, −0.56 (127)
−0.56, 0.79 (126)
0.79, 2.44 (128)
2.44, 4.20 (126)
4.20, 6.93 (129)
>= 6.93 (127)

**partial plots using CHDWB data set**




4
2
0
−2
−4
CD133 residuals
BMI residual categories
BMI residual categories (n)
< −6.10 (67)
−6.10, −4.20 (67)
−4.20, −3.22 (67)
−3.22, −1.94 (67)
−1.94, −0.80 (67)
−0.80, 0.46 (67)
0.46, 1.94 (66)
1.94, 3.58 (68)
3.58, 6.51 (67)
>= 6.51 (67)


4
2
0
−2
−4
CXCR4 residuals
BMI residual categories
BMI residual categories (n)
< −6.10 (67)
−6.10, −4.20 (67)
−4.20, −3.22 (67)
−3.22, −1.94 (67)
−1.94, −0.80 (67)
−0.80, 0.46 (67)
0.46, 1.94 (66)
1.94, 3.58 (68)
3.58, 6.51 (67)
>= 6.51 (67)